WORKSPACE / LIVE SHIFT

A little store. A lot to learn.

Brain v4 · Scenario 01 · Every action is saved for review.

THE TEAM 01

R01

Scout 01

NEEDS REVIEW

CURRENT TASK

PICK Orange

A–1–3–1 → tote

Action	Torso level
8 / 30 max	3 / 4
Waypoint	**Gripper**
tote	Empty

RECENT ACTIONS 08

01 navigate_to
arrived ✓

02 set_torso
torso positioned ✓

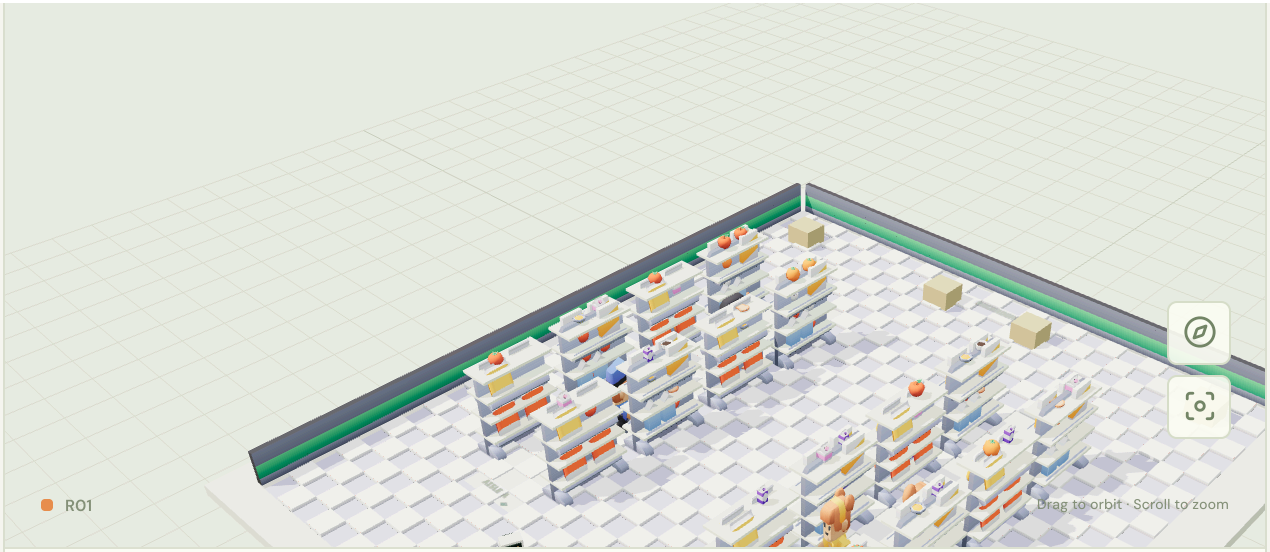
03 reach
arm extended ✓

Set up another shift

Store environment LIVE

3D view | Top view | Worker views

STORE WING A



8 / 8 actions animated

9 calls · Cache not reported

Shared brain v5

52 NODES · 70 CONNECTIONS



Knowledge every robot shares.

A pulse shows the nodes a worker read before choosing its action. Each node opens its actual brain data.

Scope · Procedure · Step · Rule · Guardrail · Protocol

The shift is ready for review.

0 of 1 tasks passed with no refused actions. 1 actions were refused.

Close review →

```
ROBOT-1 / 08   place(tote)                    item placed
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AFTER THE SHIFT / HARNESS REVIEW

Back to live view →

Every mistake has a lesson.

The supervisor connects the facts to evidence, then proposes what the shared brain should learn.

1 refused actions in the engine log · 1 tasks need review · Brain v4 · 32.3% review prompt cache

Independent evidence check

▶ Excluded findings

V2

SM

2 kept · 1 dropped

deepseek-ai/DeepSeek-V4-Pro

01 Failures 01

The moment it went wrong. Each failure opens its recorded action in the store.

F1 / ROBOT-1 ▶ STEP 4

Grasp mode must not exceed the one-hand payload limit

The store-wide default rule sets grasp.mode to one_hand for every item regardless of weight. The item (Orange, 3.98 kg) exceeds the one-hand payload limit of 2 kg, so the grasp was refused. The robot recovered with a two-hand grasp, but the refusal still fails the task.

Judge confirmed: Step 4 shows one_hand grasp refused for 3.98 kg exceeding 2 kg limit; facts match claimed check and cause.

✓ Step exists in trajectory >

02 Insights 01

What better behavior looks like, with the footage or runs behind each claim.

I1 / F1

Grasp mode must be chosen by item weight, not fixed to one_hand for all items; items over 2 kg require two_hand to avoid a refused first grasp.

Judge confirmed: Run contrasts show light items one_hand ok, heavy Orange one_hand refused then two_hand ok; supports weight-based grasp mode.

▶ Sources excluded by the judge

RUN CONTRAST REFERENCES FOUND

Failed run shift-mvcy793i/robot-1	Successful sibling shift-mvcxwrms/robot-1
1 navigate_to arrived	1 navigate_to arrived
2 set_torso torso positioned	2 set_torso torso positioned
3 reach arm extended	3 reach arm extended
4 grasp one-hand payload limit exceeded (2 kg)	4 grasp item secured
5 grasp item secured	5 retract arm retracted
6 retract arm retracted	6 navigate_to arrived
7 navigate_to arrived	7 place item placed

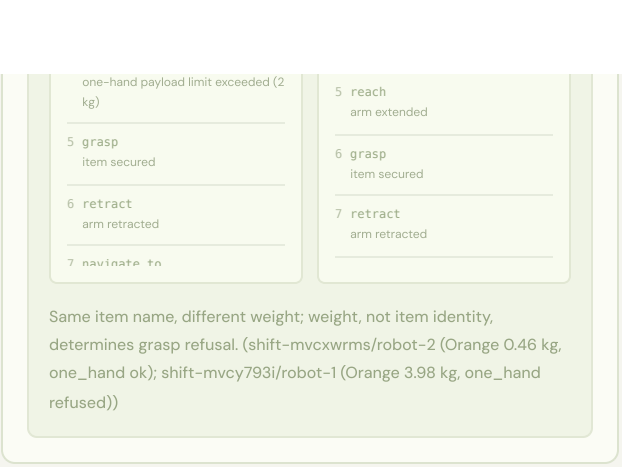
Same brain v4, same seed; the only differing factor is item weight crossing the 2 kg one-hand limit. (shift-mvcxwrms/robot-1 (Bread 0.21 kg, one_hand grasp ok, complete, 0 refused); shift-mvcy793i/robot-1 (Orange 3.98 kg, one_hand grasp refused, incomplete, 1 refused))

RUN CONTRAST REFERENCES FOUND

Failed run shift-mvcy793i/robot-1	Successful sibling shift-mvcxwrms/robot-2
1 navigate_to arrived	1 navigate_to arrived
2 set_torso torso positioned	2 set_torso torso positioned
3 reach arm extended	3 reach torso is above slot
4 grasp	4 set_torso torso positioned

V2

SM



Same item name, different weight; weight, not item identity, determines grasp refusal. (shift-mvcxwrms/robot-2 (Orange 0.46 kg, one_hand ok); shift-mvcy793i/robot-1 (Orange 3.98 kg, one_hand refused))

03 Brain updates

01

The proposed lesson in plain words, with its exact graph operations underneath.

U1 / VALID OPERATIONS

Add a weight-based grasp rule so heavy items use two_hand and avoid a refused first grasp.

Graph operations

Supervisor edits

Run post-training
VERSION 3

Adversarial attack
VERSION 3

THE NEXT SHIFT / KEEP ONLY WHAT HELPS

A change has to earn its place.

The old and candidate brains face the same ten held-out seeds, using Lightning workers on Crusoe.

Brain v5 earned the next shift.

held-out completion 60% -> 90%, refused actions per task 0.5 -> 0.2

150 MODEL CALLS

V2

SM

BEFORE / BRAIN v4

60%

6 / 10 tasks passed

0.5 refused actions · 7.5 steps per task

CANDIDATE / BRAIN v5

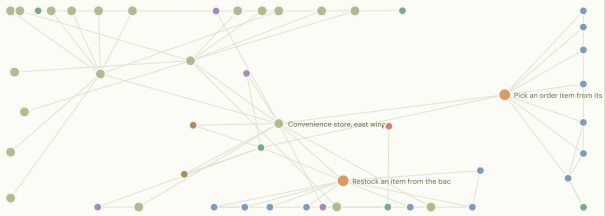
90%

9 / 10 tasks passed

0.2 refused actions · 7.2 steps per task

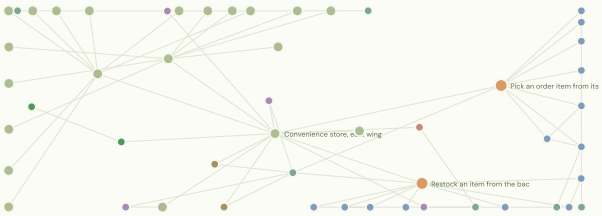
Before v4

Shared brain v4



Candidate v5

Shared brain v5

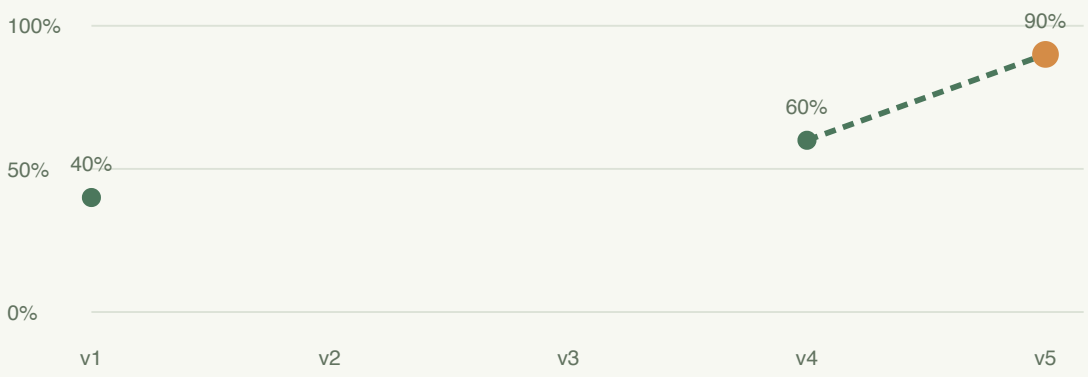


2 added 0 properties changed 0 retired

LIGHTNING EVALUATION

What the shared brain has learned

Held-out task completion by brain version. Only versions measured in the same gate are connected. Task sets may differ; unevaluated versions have no score.



v1	v2	v3

V2

SM

40.0%
superseded
4/10 tasks · 10 seeds · Lightning

Not evaluated
rejected

Not evaluated
rejected

v4
60.0%
superseded
6/10 tasks · 10 seeds · Lightning

v5 · live
90.0%
live
9/10 tasks · 10 seeds · Lightning